

Question: 01

Life expectancy at birth			Expected years of schooling			Mean years of schooling			Gross national income per capita		
Female	Male	Overall	Female	Male	Overall	Female	Male	Overall	Female	Male	Overall
84.1	79.0	81.6	16.4	15.6	16.0	12.0	12.6	12.3	43,414	64,148	53,619
80.9	77.2	78.7	16.5	15.2	15.7	12.5	12.8	12.7	28,921	77,318	62,574
80.9	74.7	77.7	16.8	15.7	16.2	12.7	13.0	12.9	30,455	47,289	38,745
82.9	77.5	80.1	20.1	20.0	20.0	11.1	11.7	11.4	22,890	35,368	29,002
80.4	72.6	76.5	16.8	15.3	16.0	13.3	13.0	13.2	25,261	41,336	33,034
80.0	77.8	78.8	17.0	15.9	16.3	10.8	11.2	11.0	16,786	53,359	39,497
78.8	68.8	73.7	16.7	15.9	16.3	13.6	13.4	13.5	33,891	42,500	37,931
81.4	76.5	78.9	17.0	16.5	16.7	10.8	11.0	10.9	17,553	31,677	24,563
80.9	78.3	79.3	14.5	12.1	12.6	11.6	9.6	10.0	42,101	104,066	87,134
78.4	71.5	74.9	15.0	14.0	14.5	12.9	13.0	12.9	24,849	36,813	30,690
77.9	71.1	74.5	15.3	14.8	15.0	12.1	12.4	12.2	25,909	40,262	32,789
77.9	70.6	74.2	14.7	13.8	14.2	11.0	11.6	11.3	24,554	35,874	30,027
75.5	68.4	71.8	14.2	13.6	13.9	11.5	11.3	11.4	18,109	28,357	23,079
79.4	75.6	77.6	17.7	13.8	15.7	10.3	9.1	9.9	10,235	14,555	12,306
81.2	75.5	78.2	14.8	13.7	14.2	7.3	7.9	7.6	13,980	20,883	17,504
76.3	69.3	72.6	15.4	13.6	14.5	9.6	9.0	9.3	13,695	22,248	17,990
73.5	64.4	68.8	14.8	14.1	14.4	11.9	11.8	11.8	12,087	17,961	14,875
76.4	71.2	73.7	15.1	13.8	14.4	12.6	12.4	12.5	5,103	10,693	7,879
74.7	70.1	72.4	15.2	15.5	15.4	9.3	10.5	9.9	9,813	14,727	12,246
81.0	79.1	79.9	14.2	11.9	12.6	7.1	7.5	7.3	6,359	22,119	15,448
72.4	68.2	70.4	11.5	12.7	12.1	7.2	8.4	7.9	2,184	17,748	9,977
73.7	69.6	71.6	11.2	12.1	11.7	10.9	11.8	11.3	2,980	6,096	4,548
74.3	67.1	70.5	13.3	12.7	13.0	9.0	8.7	8.8	4,249	8,345	6,309
76.4	71.9	74.0	13.9	14.4	14.2	5.0	6.9	5.9	3,194	11,356	7,303
75.1	66.1	70.7	12.7	12.6	12.7	6.8	7.6	7.2	5,824	11,015	8,296
66.0	61.6	63.8	12.1	12.0	12.0	7.8	9.0	8.3	4,723	6,771	5,745
72.7	66.0	69.2	10.5	10.6	10.6	5.2	6.2	5.7	4,909	12,614	8,723
72.5	67.9	70.1	10.4	9.9	10.1	6.8	7.4	7.1	4,271	6,304	5,298
70.4	65.2	67.6	13.5	13.3	13.4	5.6	6.8	6.2	2,415	5,635	4,021
63.0	55.7	59.3	11.9	11.9	11.9	7.5	6.9	7.2	7,271	10,094	8,634
62.0	56.2	59.3	12.0	12.3	12.1	8.3	9.2	8.7	3,286	4,397	3,810

- Calculate Human Development Index (HDI) of the countries and identify the most and the least developed country.
- Calculate Gender Development Index (GDI) for the countries, compare with respective HDI, and report the country having highest gender difference.
- Is there any evidence of gender difference in life expectancy at birth?

Solution

a) Calculating Human Development Index (HDI):

We have to calculate,

$$\text{Health index (HI)} = \frac{\text{life expectancy} - 20}{85 - 20}$$

$$\text{Education index (EI)} = \text{Geometric Mean} \left(\frac{EYS}{18}, \frac{MYS}{15} \right)$$

$$\text{Income index (II)} = \frac{\ln(GNI) - \ln(100)}{\ln(75000) - \ln(100)}$$

$$\text{HDI} = \text{Geometric Mean}(\text{HI}, \text{EI}, \text{II})$$

HDI Range	$\text{HDI} < 0.55$	$0.55 \leq \text{HDI} < 0.7$	$0.7 \leq \text{HDI} < 0.8$	$0.8 \leq \text{HDI}$
Comment	Low	Medium	High	Very High

Country	life index	Education index	Income index	HDI	HDI Rank	HDI Category
Austria	0.94738	0.85	0.949307	0.91524	1	Very High
UAE	0.90324	0.86	0.972637	0.91066	2	Very High
Czechia	0.88813	0.88	0.900231	0.88916	4	Very High
Greece	0.92478	0.92	0.856481	0.89985	3	Very High
Poland	0.86857	0.88	0.876142	0.87622	5	Very High
Bahrain	0.90401	0.82	0.903134	0.87359	7	Very High
Lithuania	0.82646	0.9	0.897024	0.87457	6	Very High
Chile	0.90682	0.82	0.831386	0.85298	9	Very High
Qatar	0.91187	0.68	1.022652	0.86081	8	Very High
Slovakia	0.84477	0.83	0.865026	0.84752	10	Very High
Hungary	0.83892	0.83	0.875017	0.84635	11	Very High
Romania	0.83361	0.77	0.861726	0.82118	12	Very High
Bulgaria	0.79689	0.77	0.821971	0.79478	13	High
Barbados	0.88571	0.76	0.726986	0.78708	14	High
China	0.89555	0.63	0.780208	0.76192	18	High
Dominican Republic	0.80946	0.71	0.784339	0.76536	16	High
Moldova	0.75148	0.79	0.755624	0.7671	15	High
Cuba	0.82589	0.82	0.659625	0.76371	17	High
Peru	0.8058	0.75	0.726242	0.76016	19	High
Maldives	0.92182	0.59	0.761332	0.74339	20	High
Iraq	0.77504	0.59	0.695293	0.68441	21	Medium
Tajikistan	0.79376	0.7	0.576612	0.68406	22	Medium
Belize	0.77646	0.65	0.626062	0.68197	23	Medium
Morocco	0.83141	0.56	0.648157	0.66949	25	Medium
El Salvador	0.78074	0.58	0.667412	0.67082	24	Medium
Ghana	0.67378	0.61	0.611901	0.6311	26	Medium
Guatemala	0.75749	0.47	0.675005	0.62235	27	Medium
Honduras	0.77112	0.52	0.599677	0.62015	28	Medium
Sao Tome and Principe	0.73217	0.55	0.558031	0.60969	30	Medium
Namibia	0.60414	0.56	0.673442	0.61209	29	Medium
Zimbabwe	0.60389	0.63	0.549871	0.59209	31	Medium

Human Development Index (HDI) is calculated using life expectancy, education, and income indices. Among the listed countries, **Austria** achieved the highest HDI (approximately 0.91524), indicating the highest

level of human development. **Zimbabwe** had the lowest HDI (approximately 0.59209), indicating the lowest level of development among the countries considered.

b) Calculating Gender related Development Index (GDI) with respective HDI:

Female HDI and **Male HDI** can be calculated same way. Then we can calculate the **GDI**.

$$GDI = \frac{\text{Female HDI}}{\text{Male HDI}} \text{ and}$$

$$\text{Gender diff} = \text{Female HDI} - \text{Male HDI}$$

Country	Female HDI	Male HDI	GDI	Gender Difference
Austria	0.917114	0.91114	1.006557	0.005974
UAE	0.889129	0.908985	0.978155	0.019857
Czechia	0.895748	0.879982	1.017916	0.015766
Greece	0.8965	0.901028	0.994974	0.004528
Poland	0.89071	0.858126	1.037971	0.032584
Bahrain	0.8382	0.880507	0.951951	0.042308
Lithuania	0.900449	0.847418	1.062579	0.05303
Chile	0.847618	0.85313	0.993539	0.005512
Qatar	0.877955	0.851694	1.030833	0.026261
Slovakia	0.858904	0.834066	1.02978	0.024838
Hungary	0.85244	0.837442	1.017909	0.014998
Romania	0.83065	0.810339	1.025065	0.020311
Bulgaria	0.804653	0.783235	1.027346	0.021418
Barbados	0.806295	0.759583	1.061497	0.046712
China	0.763345	0.757586	1.007601	0.005759
Dominican	0.780612	0.74669	1.045429	0.033921
Moldova	0.783113	0.749403	1.044982	0.03371
Cuba	0.756122	0.762463	0.991684	0.006341
Peru	0.750745	0.767039	0.978757	0.016294
Maldives	0.711377	0.753434	0.94418	0.042057
Iraq	0.592869	0.714597	0.829655	0.121728
Tajikistan	0.657683	0.700528	0.938839	0.042845
Belize	0.681012	0.675945	1.007497	0.005068
Morocco	0.612812	0.701538	0.873525	0.088727
El Salvador	0.664969	0.670004	0.992485	0.005035
Ghana	0.624009	0.636519	0.980346	0.01251
Guatemala	0.598936	0.63526	0.94282	0.036324
Honduras	0.615823	0.622114	0.989888	0.006291
Sao Tome and Principe	0.581954	0.625631	0.930187	0.043677
Namibia	0.626101	0.596097	1.050334	0.030004
Zimbabwe	0.591451	0.590487	1.001632	0.000964

Comment: A GDI value equal to 1 indicates perfect gender equality in human development between males and females. A GDI value less than 1 indicates female disadvantage, meaning female HDI is lower than male HDI. Conversely, a GDI value greater than 1 indicates female advantage, where female HDI exceeds male HDI.

Gender related Development Index (GDI) is calculated as the ratio of female HDI to male HDI. The results show that Iraq has the largest gender disparity (0.121728), with female HDI =0.592869, male HDI =0.714597 and GDI =0.829655, indicating substantially lower female HDI compared to male HDI. Therefore, Iraq is the country with the highest gender difference among the given countries.

c) To examine gender difference in life expectancy at birth, a paired-samples t-test was conducted using female and male life expectancy values for each country.

At first, we set up an appropriate hypothesis:

$H_0: \mu_d = 0$ (There is no gender difference)

$H_1: \mu_d \neq 0$ (H_0 is not true)

Paired Samples Test								
	Paired Differences					t	df	Sig. (2-tailed)
	Mean	Std. Deviation	Std. Error Mean	95% Confidence Interval				
				Lower	Upper			
F_HDI - M HDI	-.0039585	.0382048	.0068618	-.0179721	.0100552	-.577	30	.568

Comment: Since p-value is less than 0.05, we may reject our null hypothesis in favor of alternative hypothesis. The result shows a statistically significant difference between the two groups ($p < 0.05$). Female life expectancy is significantly higher than male life expectancy.

Question: 02

India				Malaysia			
GDP per capita, PPP (constant 2017 international \$)	School enrollment t (%) gross)	Literacy rate, adult total (% of people ages 15 and above)	Life expectancy at birth, total (years)	GDP per capita, PPP (constant 2017 international \$)	School enrollment t (%) gross)	Literacy rate, adult total (% of people ages 15 and above)	Life expectancy at birth, total (years)
2656.66	9.69	61.01	62.91	15652.01	25.00	93.02	72.78
2711.51	10.18	61.21	63.30	16155.60	27.44	93.06	72.98
2876.74	10.63	61.45	63.70	16751.41	30.61	93.10	73.17
3054.92	10.93	61.55	64.10	17538.49	30.03	93.10	73.36
3245.30	10.68	62.01	64.50	18114.14	27.92	93.10	73.56
3453.11	11.49	62.75	64.92	18752.55	28.41	93.11	73.74
3661.96	13.13	63.81	65.35	19546.99	30.01	93.12	73.93
3720.09	15.05	64.74	65.79	20103.50	33.44	93.12	74.12
3956.36	16.03	66.75	66.24	19442.99	35.49	93.12	74.31
4234.98	17.83	68.25	66.69	20536.37	37.03	93.12	74.49
4399.89	22.76	69.30	67.13	21289.26	36.15	93.22	74.68

4583.17	24.27	69.50	67.55	22132.22	37.61	93.55	74.88
4818.52	23.79	70.22	67.93	22855.95	39.07	93.93	75.07
5116.63	25.43	70.91	68.29	23906.23	39.51	94.12	75.27
5464.39	26.77	71.30	68.61	24787.82	45.59	94.55	75.46
5851.39	26.83	72.54	68.90	25541.77	46.76	94.88	75.65
6182.92	27.44	73.34	69.17	26661.51	43.72	95.08	75.83
6514.00	28.06	74.37	69.42	27577.38	45.13	94.85	76.00
6689.37	28.57	74.55	69.66	28421.46	43.06	94.97	76.16
6186.61	29.44	75.32	69.89	26471.67	42.57	95.00	76.31
6675.35	29.44	75.32	69.89	26959.25	42.57	95.00	76.31

GDP per capita, PPP (constant 2017 international \$)	Nepal			GDP per capita, PPP (constant 2017 international \$)	Thailand		
	School enrollment t (%) gross)	Literacy rate, adult total (% of people ages 15 and above)	Life expectancy at birth, total (years)		School enrollment t (%) gross)	Literacy rate, adult total (% of people ages 15 and above)	Life expectancy at birth, total (years)
2136.33	4.50	48.61	62.95	10053.82	39.07	93.12	70.83
2106.15	5.18	49.57	63.57	10583.76	40.00	93.13	71.09
2158.23	5.24	50.77	64.17	11260.16	40.95	93.22	71.39
2229.21	6.14	51.61	64.73	11886.32	42.07	92.23	71.73
2277.63	7.74	52.77	65.26	12304.40	44.58	93.51	72.10
2325.17	8.22	54.01	65.77	12837.88	44.86	93.61	72.50
2375.70	10.13	55.61	66.26	13460.06	49.03	93.69	72.92
2493.88	11.15	56.89	66.73	13620.53	48.67	93.45	73.34
2585.90	11.16	57.61	67.18	13458.53	49.40	93.89	73.77
2697.43	14.34	57.11	67.61	14399.04	50.37	96.43	74.18
2786.84	14.45	59.63	68.03	14450.48	52.26	95.43	74.59
2922.62	16.55	60.53	68.43	15424.54	50.68	95.22	74.99
3033.78	17.11	62.63	68.81	15767.36	49.85	93.70	75.38
3217.45	15.99	63.23	69.17	15854.13	50.18	93.10	75.75
3331.87	14.97	64.43	69.52	16285.39	50.58	92.87	76.09
3315.81	11.62	65.63	69.85	16782.10	49.29	93.27	76.40
3565.20	11.59	66.73	70.17	17422.95	49.89	93.67	76.68
3773.72	12.41	67.91	70.48	18101.52	49.99	93.77	76.93
3952.76	13.33	68.11	70.78	18438.65	50.29	93.88	77.15
3789.15	13.46	68.31	71.07	17253.07	50.34	93.97	77.34
3878.44	13.46	68.31	71.07	17485.51	50.34	93.97	77.34

- (a) Calculate Human Development Index (HDI) of the countries over the given period.
- (b) Construct time series plots of HDIs of the countries and compare progresses among those countries.

Solution:

a) We have to calculate,

$$\text{Health index (HI)} = \frac{\text{life expectancy} - 20}{85 - 20}$$

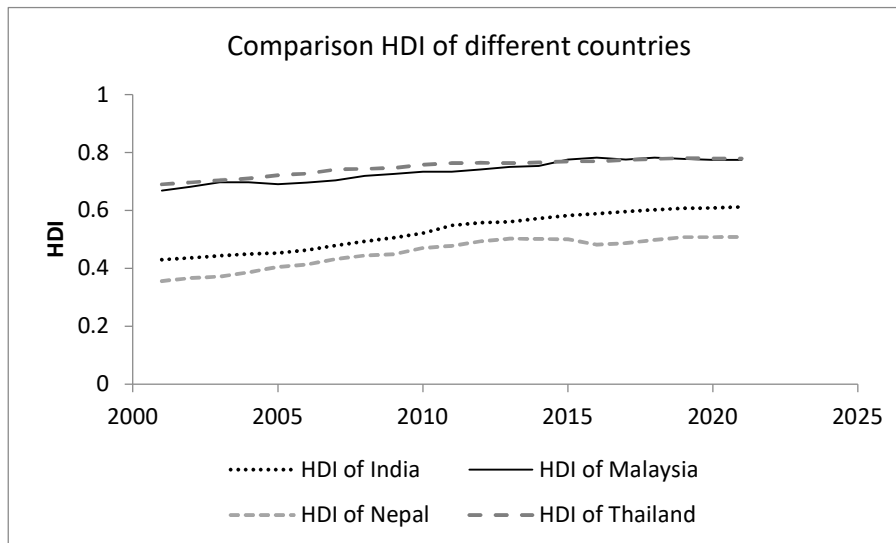
$$\text{Education index (EI)} = \text{Geometric Mean} \left(\frac{\text{School Enrollment}}{100}, \frac{\text{Literacy rate}}{100} \right)$$

$$\text{Income index (II)} = \frac{\ln(\text{GNI}) - \ln(100)}{\ln(75000) - \ln(100)}$$

$$\text{HDI} = \text{Geometric Mean}(\text{HI}, \text{EI}, \text{II})$$

Year	HDI of India	HDI of Malaysia	HDI of Nepal	HDI of Thailand
2001	0.430051	0.66860026	0.356148	0.690003
2002	0.436014	0.68138395	0.367078	0.696442
2003	0.4434	0.69642838	0.371824	0.703755
2004	0.449628	0.69715817	0.385848	0.709919
2005	0.452433	0.69097773	0.405005	0.721885
2006	0.463016	0.69534886	0.413134	0.726739
2007	0.478891	0.70438299	0.432386	0.742052
2008	0.493412	0.71932197	0.444707	0.743392
2009	0.505669	0.725814	0.448821	0.747194
2010	0.521486	0.73431845	0.470759	0.75837
2011	0.548065	0.73400513	0.477711	0.763788
2012	0.557817	0.74197018	0.493538	0.76478
2013	0.560875	0.74957348	0.50226	0.76353
2014	0.572379	0.75416564	0.501499	0.765513
2015	0.582301	0.77557312	0.500373	0.769131
2016	0.588673	0.78160017	0.482063	0.769292
2017	0.595719	0.77596627	0.487531	0.774549
2018	0.602867	0.78212697	0.498185	0.777984
2019	0.607191	0.7783284	0.507519	0.780832
2020	0.608403	0.77428711	0.507598	0.778634
2021	0.612119	0.77513058	0.50868	0.779308

(b) Time series plots of HDIs of the countries:



Comment: The graph shows the comparison of Human Development Index (HDI) among India, Malaysia, Nepal, and Thailand from 2001 to 2021. It can be observed that the HDI values of all four countries increased gradually over the years, indicating improvement in human development. This reflects better progress in education, healthcare, and standard of living in these countries.

Thailand and Malaysia maintained comparatively higher HDI values throughout the study period. Their HDI levels increased steadily and remained above the other two countries, showing stronger socio-economic development and better living conditions. Thailand achieved the highest HDI by the end of the period, followed closely by Malaysia.

India and Nepal also showed continuous improvement in HDI over time. India experienced significant growth and reduced the gap with the higher-ranked countries. Nepal, although having the lowest HDI among the four countries, demonstrated noticeable progress during the period. Overall, the graph indicates a positive trend in human development for all countries.

Question: 03

Country	Human Development Index (HDI)	Gini coefficient			
			Saint Lucia	0.715	51.2
			South Africa	0.713	63.0
			Samoa	0.707	38.7
Australia	0.951	34.3	Gabon	0.706	38.0
Denmark	0.948	27.7	Lebanon	0.706	31.8
Sweden	0.947	29.3	Indonesia	0.705	37.3
Ireland	0.945	30.6	Viet Nam	0.703	35.7
Germany	0.942	31.7	El Salvador	0.675	38.8
Netherlands	0.941	29.2	Nicaragua	0.667	46.2
Finland	0.940	27.7	Bhutan	0.666	37.4
United States	0.921	41.5	Cabo Verde	0.662	42.4
Israel	0.919	38.6	Bangladesh	0.661	32.4
Malta	0.918	31.0	Tuvalu	0.641	39.1
Slovenia	0.918	24.4	Marshall Islands	0.639	35.5
Austria	0.916	30.2	India	0.633	35.7
UAE	0.911	26.0	Ghana	0.632	43.5
Spain	0.905	34.3	Micronesia	0.628	40.1
France	0.903	32.4	Guatemala	0.627	48.3
Croatia	0.858	28.9	Kiribati	0.624	27.8
Chile	0.855	44.9	Honduras	0.621	48.2
Slovakia	0.848	23.2	Sao Tome and Principe	0.618	40.7
Hungary	0.846	30.0	Namibia	0.615	59.1
Argentina	0.842	42.3	Lao People's Democratic Republic	0.607	38.8
Türkiye	0.838	41.9	Timor-Leste	0.607	28.7
Montenegro	0.832	36.8	Vanuatu	0.607	32.3
Russian Federation	0.822	36.0	Congo	0.571	48.9
Romania	0.821	34.8	Zambia	0.565	57.1
Kazakhstan	0.811	27.8	Solomon Islands	0.564	37.1
Costa Rica	0.809	49.3	Comoros	0.558	45.3
Uruguay	0.809	40.2	Papua New Guinea	0.558	41.9
Belarus	0.808	24.4	Mauritania	0.556	32.6
Malaysia	0.803	41.1	Côte d'Ivoire	0.550	37.2
Georgia	0.802	34.5	Lesotho	0.514	44.9
Mauritius	0.802	36.8	Malawi	0.512	38.5
Serbia	0.802	34.5	Senegal	0.511	38.1
Thailand	0.800	35.0	Djibouti	0.509	41.6
Sri Lanka	0.782	39.3	Sudan	0.508	34.2
Bosnia and Herzegovina	0.780	33.0	Madagascar	0.501	42.6
Iran	0.774	40.9	Gambia	0.500	35.9
Ukraine	0.773	25.6	Ethiopia	0.498	35.0
North Macedonia	0.770	33.0	Guinea-Bissau	0.483	34.8
China	0.768	38.2	Liberia	0.481	35.3
Colombia	0.752	54.2	Congo	0.479	42.1
Maldives	0.747	29.3	Sierra Leone	0.477	35.7
Algeria	0.745	27.6	Guinea	0.465	29.6
Tonga	0.745	37.6	Yemen	0.455	36.7
Ecuador	0.740	47.3	Burkina Faso	0.449	47.3
Mongolia	0.739	32.7	Mozambique	0.446	54.0
Egypt	0.731	31.5	Mali	0.428	36.1
Tunisia	0.731	32.8	Burundi	0.426	38.6
Jordan	0.720	33.7			
Paraguay	0.717	43.5			
Palestine	0.715	33.7			

- (a) Compute rank correlations between the HDI and Gini coefficient and comment on the direction and strength of relationship between development and inequality.
- (b) Compute average inequality measures for different levels of development. Do they differ significantly?

Solution

- (a) Rank correlations between the HDI and Gini coefficient:

The Spearman rank correlation formula is:

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$$

where,

d_i = difference between paired ranks

n = number of observations

By using SPSS,

Correlations				HDI	Gini
Spearman's rho	HDI	Correlation Coefficient		1.000	-.384**
		Sig. (2-tailed)		.	.000
		N		100	100
	Gini	Correlation Coefficient		-.384**	1.000
		Sig. (2-tailed)		.000	.
		N		100	100

** . Correlation is significant at the 0.01 level (2-tailed).

Comment: The correlation coefficient is negative ($r_s = -0.384$), indicating an inverse/negative relationship between HDI and income inequality. This means that countries with higher HDI generally tend to have lower Gini coefficients (lower inequality). The magnitude (strength and direction) of the correlation indicates a weak to moderate negative relationship.

Since the p-value is less than 0.01, there exists a statistically significant negative relationship between human development and income inequality. Countries with higher HDI tend to exhibit lower levels of inequality, although the strength of the relationship is moderate.

- (b) Average inequality measures for different levels of development:

By using SPSS,

Descriptives								
Gini								
	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
Very High	33	33.6758	6.32890	1.10172	31.4316	35.9199	23.20	49.30
High	24	37.9000	8.73563	1.78315	34.2113	41.5887	25.60	63.00
Medium	25	40.6040	7.81524	1.56305	37.3780	43.8300	27.80	59.10
Low	18	38.9444	5.69894	1.34325	36.1104	41.7785	29.60	54.00
Total	100	37.3700	7.66608	.76661	35.8489	38.8911	23.20	63.00

The results indicate that countries with Very High HDI have comparatively lower inequality, whereas Medium HDI countries exhibit the highest average inequality.

To determine whether these mean differences are statistically significant, a one-way ANOVA is performed.

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_1 : H_0 \text{ is not true}$$

ANOVA

Gini					
	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	763.195	3	254.398	4.831	.004
Within Groups	5054.915	96	52.655		
Total	5818.110	99			

Comment: A one-way ANOVA shows a statistically significant difference in mean Gini coefficients across different HDI groups ($F = 4.83$; $p = 0.004$). Since ($p < 0.05$), the null hypothesis of equal mean inequality was rejected, indicating at least one equality doesn't hold. This means that income inequality differs significantly among development levels.

Therefore, we have to perform multiple comparisons test by Post-hoc Tukey HSD analysis.

Post Hoc Tests

Multiple Comparisons

Dependent Variable: Gini							
	(I) Development	(J) Development	Mean Difference (I-J)	Std. Error	Sig.	95% Confidence Interval	
	Level	Level				Lower Bound	Upper Bound
Tukey HSD	Very High	High	-4.22424	1.94669	.139	-9.3141	.8656
		Medium	-6.92824*	1.92401	.003	-11.9588	-1.8977
		Low	-5.26869	2.12624	.070	-10.8280	.2906
	High	Very High	4.22424	1.94669	.139	-.8656	9.3141
		Medium	-2.70400	2.07369	.563	-8.1259	2.7179
		Low	-1.04444	2.26258	.967	-6.9602	4.8713
	Medium	Very High	6.92824*	1.92401	.003	1.8977	11.9588
		High	2.70400	2.07369	.563	-2.7179	8.1259
		Low	1.65956	2.24310	.881	-4.2053	7.5244
	Low	Very High	5.26869	2.12624	.070	-.2906	10.8280
		High	1.04444	2.26258	.967	-4.8713	6.9602
		Medium	-1.65956	2.24310	.881	-7.5244	4.2053

LSD	Very High	High	-4.22424*	1.94669	.032	-8.0884	-.3601
		Medium	-6.92824*	1.92401	.001	-10.7474	-3.1091
		Low	-5.26869*	2.12624	.015	-9.4892	-1.0481
	High	Very High	4.22424*	1.94669	.032	.3601	8.0884
		Medium	-2.70400	2.07369	.195	-6.8202	1.4122
		Low	-1.04444	2.26258	.645	-5.5356	3.4467
	Medium	Very High	6.92824*	1.92401	.001	3.1091	10.7474
		High	2.70400	2.07369	.195	-1.4122	6.8202
		Low	1.65956	2.24310	.461	-2.7930	6.1121
	Low	Very High	5.26869*	2.12624	.015	1.0481	9.4892
		High	1.04444	2.26258	.645	-3.4467	5.5356
		Medium	-1.65956	2.24310	.461	-6.1121	2.7930

*. The mean difference is significant at the 0.05 level.

Multiple Comparisons Dependent Variable: Gini

Comment: Tukey's finds Very High vs Medium showing a significant mean difference, meaning only very high development countries have significantly lower Gini coefficient than medium. All other pairwise differences are statistically insignificant.

LSD comparison finds two significant differences, Very High vs Medium and High vs Very High.

Question: 04

Country Name	Immunization, DPT (% of children ages 12- 23 months)	Pregnant women receiving prenatal care (%)	Mortality rate, under-5 (per 1,000 live births)
Afghanistan	66	65.2	58
Albania	98	88.4	9.8
Algeria	91	95.3	22.7
Armenia	93	99.6	10.9
Azerbaijan	89	91.7	19.4
Bangladesh	98	75.2	29.1
Belarus	98	99.9	2.9
Bhutan	98	97.9	27.6
Bosnia and Herzegovina	73	87	5.7
Brazil	68	97.2	14.7
Cambodia	92	95.3	25.7
Central African Republic	42	51.8	103
Chad	58	54.7	110
Colombia	86	97.2	13.2
Congo, Dem. Rep.	65	82.4	81.2
Congo, Rep.	77	93.5	44.6
Costa Rica	99	97.6	7.9
Cote d'Ivoire	76	93.2	77.9
Ethiopia	65	73.6	48.7
Gambia, The	82	97.8	49.4
Georgia	85	97.6	9.3
Ghana	98	97.4	44.7
Guatemala	79	91.3	23.6
Guinea-Bissau	67	97	76.8
Iran, Islamic Rep.	98	96.9	12.9
Iraq	78	87.6	25.2

- Compute bivariate correlations between the indicators and comment.
- Obtain the best set of predictors for UHC service coverage index.
- Interpret the contributions of predictors on UHC service coverage index.

Solution:

Correlations				
		DPT	Prenatal	Mortality
DPT	Pearson Correlation	1	.698**	-.745**
	Sig. (2-tailed)		.000	.000
	N	26	26	26
Prenatal	Pearson Correlation	.698**	1	-.677**
	Sig. (2-tailed)	.000		.000
	N	26	26	26
Mortality	Pearson Correlation	-.745**	-.677**	1
	Sig. (2-tailed)	.000	.000	
	N	26	26	26

** . Correlation is significant at the 0.01 level (2-tailed).

Comment:

Pearson's bivariate correlation analysis was conducted to examine the relationships among DPT immunization coverage, prenatal care coverage, and under-5 mortality rate.

The results shows that DPT immunization and prenatal care have a strong positive correlation ($r=0.698$, $p<0.01$). This indicates that countries with higher immunization coverage tend to provide better prenatal healthcare services.

DPT immunization was found to have a strong negative correlation with under-5 mortality rate ($r = -0.745$, $p<0.01$). This suggests that higher immunization coverage is associated with lower child mortality.

Similarly, prenatal care also showed a significant negative correlation with mortality rate ($r=-0.677$, $p<0.01$), implying that improved maternal healthcare contributes to reducing under-5 deaths.

All correlations were statistically significant at the 1% level. Overall, the findings indicate that better healthcare services are strongly associated with lower child mortality and improved health outcomes.

(b) Obtain the best set of predictors for UHC service coverage index:

To identify the best predictors of the UHC service coverage index, multiple linear regression analysis was performed using DPT immunization coverage, prenatal care coverage, and under-5 mortality rate as explanatory variables.

The regression model is:

$$UHC = \beta_0 + \beta_1 \times DPT + \beta_2 \times Prenatal + \beta_3 \times Mortality + \epsilon$$

where,

DPT = Immunization coverage

Prenatal = Prenatal care coverage

Mortality = Under-5 mortality rate

A stepwise multiple regression method was used to determine the best subset of predictors.

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Change Statistics				
					R Square Change	F Change	df1	df2	Sig. F Change
1	.933 ^a	.870	.864	.03798	.870	160.324	1	24	.000
2	.998 ^b	.996	.995	.00720	.126	644.342	1	23	.000
3	1.000 ^c	.999	.999	.00283	.004	126.939	1	22	.000

a. Predictors: (Constant), DPT

b. Predictors: (Constant), DPT, Prenatal

c. Predictors: (Constant), DPT, Prenatal, Mortality

Comment: The model summary shows that the predictive ability of the regression model improves as more predictors are included. DPT immunization alone explains 87.0% of the variation in UHC service coverage ($R^2=0.870$). Adding prenatal care increases the explained variation to 99.6%, while the final model including DPT, prenatal care, and mortality explains 99.9% of the variation ($R^2=0.999$). Therefore, these three variables together provide the best set of predictors for UHC service coverage index.

This is obvious as the UHC index is derived from those indices.

Coefficients ^a								
Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	.368	.041		8.890	.000	.282	.453
	DPT	.632	.050	.933	12.662	.000	.529	.735
2	(Constant)	.225	.010		23.314	.000	.205	.245
	DPT	.398	.013	.587	30.093	.000	.370	.425
	Prenatal	.377	.015	.495	25.384	.000	.346	.407
3	(Constant)	-.043	.024		-1.784	.088	-.093	.007
	DPT	.362	.006	.535	59.707	.000	.350	.375
	Prenatal	.354	.006	.465	57.323	.000	.341	.367
	Mortality	.329	.029	.098	11.267	.000	.268	.390

a. Dependent Variable: UHC

The fitted model is:

$$\text{UHC} = 0.535(\text{DPT}) + 0.465(\text{Prenatal}) + 0.098(\text{Mortality})$$

(c) Comment: The coefficients table indicates that all predictors make significant contributions to the UHC service coverage index ($p < 0.001$). In the final model, DPT immunization coverage ($\beta_1 = 0.535$) and prenatal care coverage ($\beta_2 = 0.465$) have positive effects on UHC service coverage, meaning that higher immunization and prenatal care are associated with better healthcare coverage. In contrast, under-5 mortality rate has a negative effect ($\beta_3 = -0.098$), indicating that higher mortality is associated with lower UHC coverage. Among the predictors, **DPT immunization** appears to be the strongest predictor because it has the highest standardized beta coefficient. The VIF values are all below 5, suggesting that multicollinearity is not a serious problem in the model.